Diffusion Prior-Based Amortized Variational Inference for Noisy Inverse Problems (DAVI)

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01 Noisy Inverse Problems: Setup and Challenge

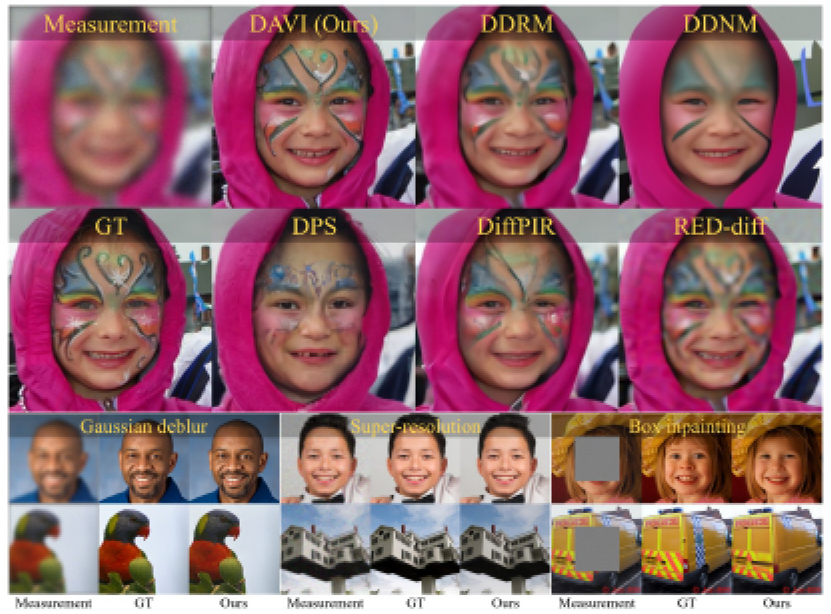
- Goal: recover a clean signal from a noisy measurement given a known linear operator and Gaussian noise
- Measurement likelihood is Gaussian around the transformed clean signal; prior knowledge is essential
- Ill-posedness: many solutions explain the same measurement, demanding strong image priors
- Applications span image restoration, medical imaging, and astronomy
- Key challenge: balancing data fidelity and prior realism under noise

02 Limits of Iterative Diffusion Inference

- Diffusion priors are strong, but iterative sampling is slow and resource heavy
- Per-sample optimization hurts scalability and generalization to new measurements
- Existing methods modify reverse diffusion or rely on spectral decompositions, adding complexity
- **Desired**: amortized, single-step inference that transfers across tasks
- Bottleneck: many network evaluations per sample and sensitivity to noise settings

[Jordan et al. 1999; Song et al. 2023]

03 DAVI Overview, Contributions, and Teaser



- **Core idea**: learn an implicit posterior that maps a measurement to samples in one step
- Optimize the KL objective using a diffusion prior with an analytic likelihood term
- Introduce Perturbed Posterior Bridge to enhance generalization
- Outcome: single-step, high-quality restoration across tasks
- Contributions: amortized variational inference, integral KL, and a bridge for robust training

04 Diffusion Priors and (Amortized) VI

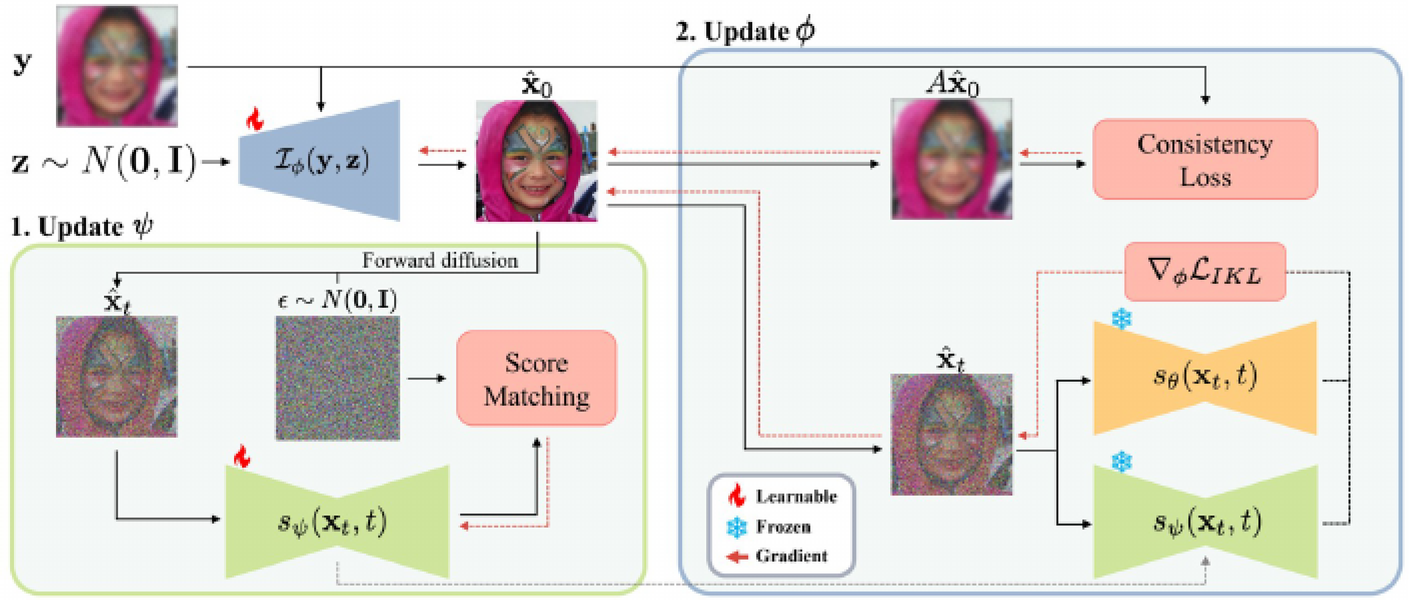
- Diffusion priors for inverse problems: DDRM, DDNM+, PiGDM, DPS, FPS, RED-diff, DiffPIR
- Variational inference approximates posteriors; amortized VI shares computation across samples
- Prior approaches rely on multi-step reverse trajectories or spectral tools
- DAVI targets **single-step** inference instead of multi-step trajectories
- Efficiency gain: one network evaluation per sample

05 Diffusion SDEs, Scores, and Notation

- Forward diffusion gradually corrupts data; the reverse process requires the data score
- A score network is trained with denoising score matching to estimate the gradient of log density
- Standard kernels mix clean data with Gaussian noise to produce perturbed variables over time
- We use a pre-trained score as a powerful prior for natural images
- Notation: clean sample at start time, standard normal at large time, and a time-weighted loss

[Anderson et al. 1982]

06 Bridge to DAVI: Using sθ as Prior + Likelihood



- Approximate the posterior using the measurement likelihood and a diffusion prior
- Alternate training of an implicit score and a measurement-to-sample mapper
- Minimize the KL between the implicit and true posteriors across many measurements
- Strategy: optimize a unified objective that couples data and prior information

07 Method

DAVI Formulation: Implicit Posterior and Objective

Implicit posterior: a neural mapper takes a noise-perturbed measurement and outputs a sample

Use the reparameterization trick with Gaussian noise for single-step sampling

Optimize the KL divergence between the implicit and true posteriors

Zero-shot: the learned mapper generalizes to unseen measurements without test-time optimization

Objective Decomposition

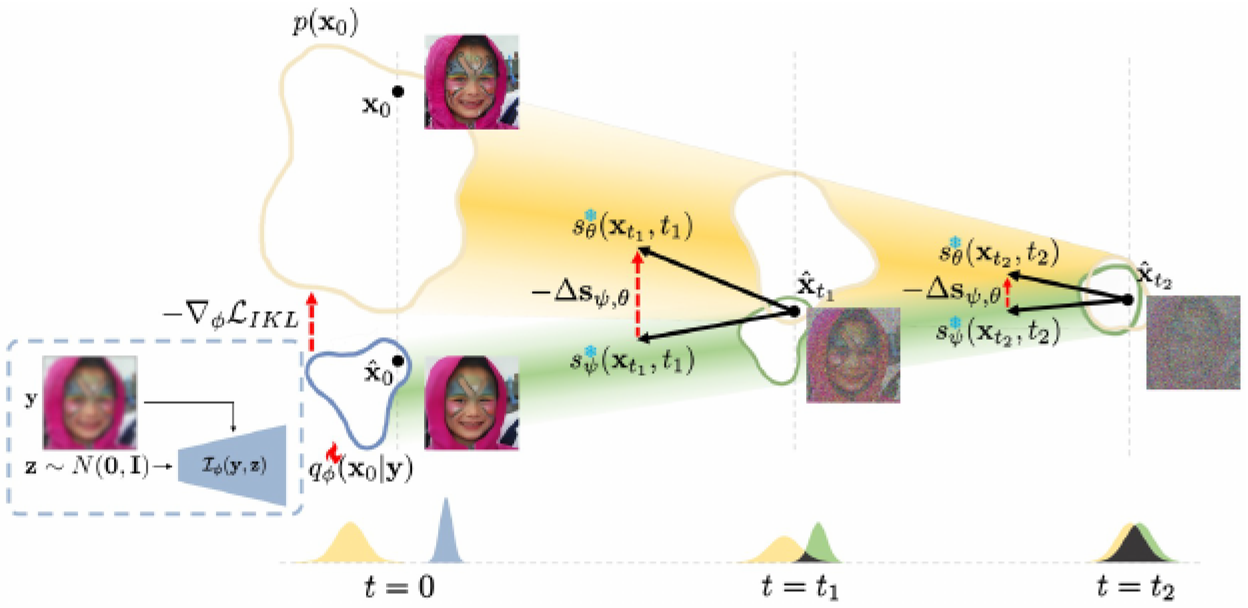
KL splits into a data consistency term and a prior matching term

Data consistency is analytic under Gaussian noise and enforces measurement fidelity

Prior matching aligns the implicit posterior with the diffusion prior

Benefit: clear training signal from both likelihood and prior for stable learning

08 Integral KL (IKL): Stable Prior Matching

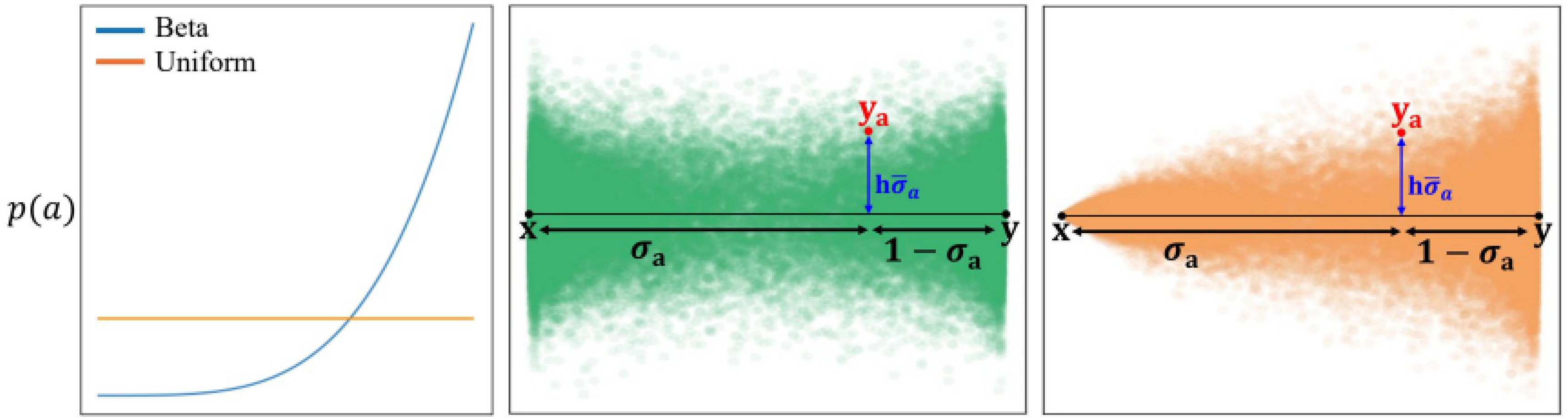


- Upper bound prior KL with an integral over time of smoothed KL terms with a positive weight
- Smoothing via the forward process mitigates support mismatch and instability
- As perturbation grows, distributions overlap more, improving robustness
- Benefit: more robust convergence than direct prior KL

09 Learning with IKL: Gradients and Alternation

- The IKL gradient depends on scores of both the diffusion prior and the implicit posterior
- Introduce an implicit score network to approximate the implicit posterior score
- Alternate training: update the implicit score and the mapper in turn
- This enables practical estimation of gradients for the IKL objective
- Outcome: stable optimization that preserves fidelity and realism

10 Perturbed Posterior Bridge (PPB): Intuition and Definition



(A) Distribution of a (B) Constant $\bar{\sigma}_a$ (C) Monotonically increasing $\bar{\sigma}_a$

- Create a bridge between measurement and clean signal using a mixing variable and a schedule
- Add a controlled perturbation that grows near the measurement to aid generalization
- Sampling across the bridge yields diverse, informative training targets
- Monotonic increase of perturbation toward the measurement improves robustness

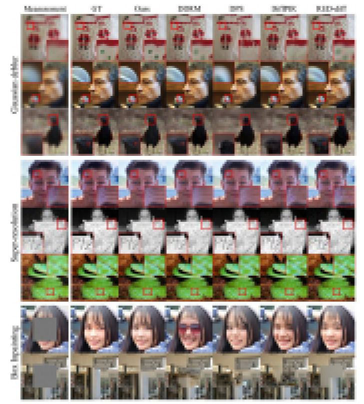
11 PPB-Conditioned Implicit Model and Inference

- Condition the mapper on the bridge variable using sinusoidal embeddings and sample it during training
- At test time set the bridge variable to one and perform a single forward pass
- Training with diverse bridge samples improves zero-shot performance
- Result: one-shot posterior sampling for unseen measurements

12 Setup: Baselines, Tasks, Metrics

- Datasets: FFHQ one thousand and ImageNet one thousand at resolution two hundred fifty six
- Tasks: Gaussian deblurring, four times super-resolution, and box inpainting with centered mask
- Noise level set to five hundredths unless stated; same unconditional diffusion prior across methods
- Metrics: PSNR, LPIPS, FID; efficiency via number of function evaluations and wall-clock time
- Implementation: shared pre-trained prior; consistent architectures for mapper and implicit score

13 Headline Quantitative and Qualitative



Method	NFE ↓	Gaussian deblur PSNR ↑	Gaussian deblur LPIPS ↓	Gaussian deblur FID ↓	4× Super-resolution PSNR ↑	4× Super-resolution LPIPS ↓	4× Super-resolution FID ↓	Box Inpainting PSNR ↑	Box Inpainting LPIPS ↓	Box Inpainting FID ↓
DDRM [18]	20	26.26	0.269	56.82	28.09	0.228	48.19	22.27	0.177	37.62
DDNM$^+$ [39]	100	24.19	0.348	91.48	28.17	0.244	59.09	23.79	0.218	48.82
DPS [4]	1000	21.88	0.315	34.47	25.55	0.255	36.29	21.94	0.302	42.45
ΠGDM [32]	100	23.05	0.320	52.72	27.73	0.225	49.86	23.04	0.220	32.99
DiffPIR [45]	100	24.41	0.299	33.91	25.32	0.341	40.35	23.97	0.195	31.27
RED-diff [23]	1000	**26.44**	0.324	46.55	26.75	0.379	92.82	24.16	0.216	35.80
DAVI (Ours)	**1**	25.46	**0.225**	**29.92**	**28.23**	**0.171**	**23.96**	**26.25**	**0.112**	**14.31**

- DAVI attains the best FID and LPIPS across tasks with a single step
- Qualitative results show sharper, realistic details while preserving measurement consistency
- Consistent gains on FFHQ and ImageNet with comparable or better PSNR

14 Robustness and Speed

Method	**DAVI** (Ours)	RED-diff [23]	DPS [4]	ΠGDM [32]	DDRM [18]	DDNM$^+$ [39]
Noise scale $\sigma_{\mathbf{y}}$	-	-	-	Required	Required	Required
PSNR ↑	**23.47**	23.25	19.81	21.80	16.09	19.25
LPIPS ↓	**0.347**	0.470	0.447	0.446	0.595	0.543
FID ↓	**63.37**	111.08	64.00	82.98	157.38	147.30

- Unknown noise scale: DAVI remains robust without requiring the noise parameter
- Inference time around four hundredths of a second per image, far faster than iterative baselines
- Practicality: suitable for real-time and resource-limited settings
- Maintains strong FID under variable noise while alternatives degrade

15 IKL and PPB Contributions

Config.	IKL	PPB	$\bar{\sigma}_a$	h	$p(a)$	PSNR ↑	LPIPS ↓	FID ↓
(A)	✗	✗	-	-	-	21.47	0.443	50.69
(B)	✓	✗	-	-	-	24.70	0.236	37.61
(C)	✓	✓	Const.	0	Uniform	**26.71**	0.247	49.37
(D)	✓	✓	Const.	0.01	Uniform	26.17	0.239	47.41
(E)	✓	✓	Const.	0.01	Beta	25.99	0.232	34.34
(F)	✓	✓	M.I	0.01	Beta	25.46	**0.225**	**29.92**

- Integral KL markedly improves realism and stability versus a direct prior KL
- Perturbed Posterior Bridge with increasing perturbation and beta sampling performs best
- Removing perturbation collapses diversity and harms perceptual quality
- Takeaway: both IKL and PPB are critical for high fidelity and generalization

16 Takeaways

- Amortized, single-step posterior sampling for noisy inverse problems
- Integral KL and Perturbed Posterior Bridge enable stable and generalizable training
- Outcome: strong fidelity and realism with large efficiency gains
- Ready for deployment on commodity hardware without test-time optimization

THANKS!